Erik Henrique Alves Cunha

Senior Software Engineer, AI Engineer | LLM Systems, Agents, Evals, Guardrails, MCP, Python, TypeScript, Node.js

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SUMMARY

8+ years of production software, focused on GenAI and AI-native products: **LLM systems**, agents and evals, on TypeScript, Python, **React, Next.js**, Angular and Node.js, with PostgreSQL, Docker and AWS. Payments, business insurance and hospital equipment service taught me audit trails and failure isolation, the same problems agent systems meet as guardrails.

I build LLM systems and the evaluation that decides whether they are safe to ship: on my own site a live LLM endpoint runs behind a continuous integration (CI) harness, with an LLM judge calibrated against my own labels at a 0.85 agreement floor and jailbreak resistance gated at 1.0. I work daily in Claude Code and Codex, authoring the skills, guard policies and lifecycle hooks they run under. At Raylu I built the table engine behind a private-markets product's data-dense pages from zero, and the migration was measured by comparing production builds of each route against its parent commit. At Betsson I built a **12-agent AI-assisted engineering system** on a PCI-DSS payment platform processing **40M+ transactions a year**.

PROFESSIONAL EXPERIENCE

Senior Software Engineer

06/2026 – Present

Raylu, *New York, US (Remote)*

- AI deal origination platform for private markets, used by 50+ venture, growth and private equity funds holding $500B+ in assets to go from an investment thesis to booked founder meetings, scanning 300,000+ sources. $8M Series A, security-audited (SOC 2 Type I and II).
- Rebuild the investor workflow screens of a private-markets platform (company, pipeline, comparison, outreach), giving funds holding **$500B+ in assets** one surface to shortlist, compare and contact founders, in React and Next.js 14 with TypeScript 5, JavaScript, HTML and CSS; TanStack; Plate, Tailwind CSS 4 and Radix; Node.js 24, Fastify, TypeORM and PostgreSQL; Nx and Storybook.
- Deliver backend changes on Fastify, TypeORM and PostgreSQL: a hand-written migration with a rehearsed backfill and tested rollback, and an EXISTS-subquery fix that raised one owner's team-filter matches from **14 to the correct 75**.
- Design the **versioned view-state contract** those data-dense pages share, encoding sort, filters, search, columns and grouping in URL-safe TypeScript codecs, with a backward-compatible migration chain, so every shared or stale link resolves to a working view.
- Measure the data-platform migration by comparing production builds of four Next.js routes against their parent commits, cutting controls without an accessible name to 1 on every route, from 55 on the worst, and on the pipelines route Cumulative Layout Shift (CLS) from **0.166 to zero**, Largest Contentful Paint (LCP) from 2,544 to 1,308 ms and Lighthouse performance from 79 to 96.
- Engineer the table engine, client mode on TanStack Table and server mode on React Query with cancellable requests, deterministic caching and faceted filter counts, to **94.7% line and 85.4% branch coverage** against library baselines of 85.4% and 65.8%.
- Specify the grid's accessibility in design documents before code (WAI-ARIA grid and row headers, roving-tabindex toolbars, live-region sort announcements), and virtualize rows over a real table element.
- Build the shared outreach editor on Plate and React, holding personalization as document structure so authors see which fields resolve before sending.
- Write the rich-text-editor design doc as a **1,050-line** engineering document whose evidence log cites code at path:line, answering every review thread it drew.
- Give **189 code reviews** across nine authors, with teammates extending the delivered data-table core in later epics.
- **Stack:** *React, Next.js 14, TypeScript, TanStack, Plate, Fastify, TypeORM, PostgreSQL, Tailwind CSS, Radix, Nx, Storybook, Chromatic, Jest, React Testing Library, Playwright.*

Senior Software Engineer

03/2025 – 05/2026

Betsson Group, *Malta (Hybrid)*

- Payment and cashier platform on the operator's proprietary Player Account Management system, under card-industry security audit (PCI-DSS), at a Nasdaq Stockholm-listed operator: €1.197B revenue and 31.1M registered customers in 2025, local licences in 24 countries, about 2,900 staff across 20 countries.
- Built the cashier micro-frontend platform every deposit and payout across the group's brands passes through, on **40M+ transactions a year**, in Angular 19 with TypeScript, JavaScript, HTML, CSS and SCSS; RxJS, NgRx; StencilJS Web Components, Storybook; Nx.
- Isolated brand orchestration from the shared payment primitives in the Nx monorepo, so each brand changed without touching payment code.
- Built a **12-agent AI-assisted engineering system** on GitHub Copilot for code generation, review, debugging, testing, and architectural validation.
- Designed the deposit and withdrawal flows as async state machines in RxJS and NgRx, with recovery paths for failed steps, across **15+ regulated jurisdictions**.
- Implemented the shared payment primitives as StencilJS Web Components driven by white-label config, rendering the right frontend per country, jurisdiction and product across **5+ brands**, 8+ languages and 10+ currencies, zero-decimal ones too.

- Integrated Trustly open-banking deposits into the cashier for the Nordic markets, as a StencilJS component configured per country and brand.
- Integrated Yape for Peru (about 16M monthly users): phone number plus app approval code, status polling and handling for timed-out and declined payments, with both providers mocked in WireMock and Playwright.
- Tested the KYC and payment flows with Jest, Playwright and WireMock against mocked providers, so provider outages and API changes did not break the suite.
- Enforced path-scoped engineering standards through Husky Git hooks, automated tooling and secret scanning.
- **Stack:** *TypeScript, Angular, RxJS, NgRx, StencilJS, Web Components, Micro-frontends, Nx Monorepo, Storybook, Jest, Playwright, WireMock, GitHub Actions, GitHub Copilot.*

Senior Software Engineer *03/2023 – 03/2025*

Forward Deployed Engineer, *Remote*

- Forward-deployed engineering for a medical imaging equipment company: the service-order and field-service platform its engineers maintain imaging equipment at hospitals through, with specialist scheduling, field visits and parts collection.
- Rebuilt a medical imaging equipment company's hospital equipment service platform, replacing a legacy tool with a layered domain, data and presentation architecture for service orders, specialist schedules and field visits, as a progressive web app on Angular 17 to 19 with TypeScript 5, JavaScript, HTML, CSS and SCSS; RxJS; PrimeNG, Chart.js, FullCalendar, Bryntum Scheduler and the Google Maps API; WebSocket (STOMP/SockJS); Nx.
- Built specialist visit scheduling with travel-time checks, plus a **29-permission** security-profile model and an HTTP interceptor that retries with backoff and de-duplicates in-flight requests.
- Automated releases with Bash, SSH and NGINX, cutting production updates from **about 5 minutes to about 45 seconds** and rollbacks to about 1 minute.
- Cut time to interactive **about 52%** with concurrent RxJS loading and change-detection tuning, and the bundle about 33%, layout-shell CSS about 98% and initial load about 25% by lazy loading and PrimeFlex utilities.
- **Stack:** *Angular, TypeScript, RxJS, Nx Monorepo, WebSocket (STOMP/SockJS), PrimeNG, Chart.js, FullCalendar, Bryntum Scheduler, Google Maps API, PWA, Husky, Bash, SSH, NGINX, GitHub Actions.*

Software Engineer *12/2021 – 03/2023*

Grupo SBF (Nike Brazil, Centauro), *São Paulo, Brazil (Remote)*

- B3-listed owner of Centauro and of Fisia, Nike's exclusive Brazil distributor through 2034: R$7.9B gross revenue in 2022 with 31.9% of sales digital, and group storefronts holding 21% of Brazil's online fashion, apparel and footwear traffic.
- Rebuilt Nike Brazil's storefront behind a BFF gateway, on the commerce platform Grupo SBF shares with Centauro, cut over in a phased rollout, in React and Next.js 12 to 13 (SSR, SSG and ISR) with TypeScript, JavaScript, HTML and CSS; styled-components; NextAuth; Nx, Webpack and Storybook.
- Built the product listing, product detail and search pages on Next.js, with filters, sorting, size grids and stock availability.
- Kept the Nike and Centauro storefronts, serving **8M+ monthly users**, isolated inside the shared Nx monorepo with enforced module boundaries, without forking the codebase.
- Cut page load **about 32%** on storefronts carrying R$2.5B of 2022 digital revenue through Next.js code splitting, dynamic imports, image optimization and caching, with Core Web Vitals about 12% better, led by LCP.
- Lifted conversion **about 10%** through 20+ Google Optimize 360 A/B experiments, each with pre-agreed stop-loss criteria.
- Built the design system shared by the Nike and Centauro storefronts, with brand theming and Storybook.
- Implemented technical SEO on the Next.js storefronts: schema.org structured data, meta tags, sitemaps, canonical URLs and incremental static regeneration for product pages.
- Instrumented e-commerce analytics through data layer events (view_item, add_to_cart, purchase), feeding the conversion funnel and the A/B experiments.
- Covered the four revenue-critical checkout journeys with integration tests in Jest, React Testing Library and MSW, at **about 70%** code coverage.
- Integrated a WebAR try-on into the Next.js purchase journey.
- Mentored four junior engineers on React patterns, testing and code review.
- **Stack:** *React, Next.js, TypeScript, styled-components (SSR extraction), NextAuth, Sentry, Nx Monorepo, Webpack, SSR/SSG, ISR, BFF gateway, Storybook, Jest, React Testing Library, MSW, Google Optimize, Google Analytics.*

Frontend Developer · assigned to VMware *01/2021 – 12/2021*

Encora (formerly Daitan), assigned to VMware, *Campinas, Brazil*

- VMware Pathfinder, the self-service portal fronting VMware's Hands-on Labs, Test Drives and assessment tools across Tanzu, Anywhere Workspace and VMware Cloud on AWS: a catalogue of live interactive labs, retired after the Broadcom acquisition. Delivered from Daitan, acquired by Encora in July 2021.
- Engineered VMware Pathfinder, the evaluation portal behind **2.1M+ labs**, where customers run live interactive labs in the browser and follow guided paths before buying, in Angular 11 with TypeScript, JavaScript, HTML, CSS and SCSS; RxJS, NgRx, Reactive Forms and i18n.
- Fed catalogue pages from CMS content, so copy changed without a deploy, localized the portal through i18n, and instrumented funnel analytics for lab starts, completions and assessments.
- Held unit-test coverage at **about 70%** with Karma and Jasmine on the workflows that gate a lab launch.
- **Stack:** *Angular, TypeScript, RxJS, NgRx, Reactive Forms, i18n, Karma, Jasmine, AWS CodeBuild, S3, CloudFront.*

Frontend Developer *04/2020 – 01/2021*
Zup Innovation (Itaú Unibanco), *Campinas, Brazil*

- Itaú Unibanco's in-house engineering company, acquired in 2019 and 1,994 strong by the end of 2020, building for Latin America's largest bank: 24.2M digital clients at the end of 2020, 1.2M of them businesses, selling partner insurers' policies through an open platform.
- Built **micro-frontends** to replace Itaú's legacy business insurance application journey by journey, packaged as custom elements and rendered at runtime by the bank's internal host platform, in Angular 9 and Angular Elements with TypeScript, JavaScript, HTML, CSS and SCSS; RxJS and Reactive Forms.
- Shipped each micro-frontend through its own AWS CodePipeline and CodeBuild pipeline to S3 and CloudFront, so it deployed on its own schedule.
- Defined each micro-frontend's contract with the host platform (custom-element attributes in, DOM CustomEvents out) and drove each insurer's flow from configuration, so a new partner insurer needed no frontend code.
- Built the offer showcase comparing plans from **4+ partner insurers** (e.g. MetLife, Amil), plus quote forms validating CNPJ and covered lives.
- **Stack:** *TypeScript, Angular, Angular Elements, RxJS, Reactive Forms, Jasmine, AWS CodePipeline, CodeBuild, S3, CloudFront.*

Full Stack Engineer → Frontend Developer *02/2019 – 04/2020*
Venturus, *Campinas, Brazil*

- Non-profit Brazilian R&D institute born inside Ericsson in 1995, one of the country's largest private research and innovation centres, with Motorola and Sony among its clients.
- Built CGO Assessoria's customs-compliance platform end to end, cutting its reporting API from **about 40s to under 1s** and other endpoints from 2s to 200ms, in Angular 8 with TypeScript, JavaScript, HTML, CSS and SCSS; RxJS; Node.js, Express, PostgreSQL and MongoDB; JWT.
- Built the Express REST API with input validation, central error middleware and paginated listings, with multi-step writes in PostgreSQL transactions and API integration tests.
- Modelled each foreign-trade process as a state machine (import licence, MAPA consent, release), with a document checklist per process type and deadline alerts.
- Traced the slow reporting endpoint to how the ORM queried the data and fixed it with rewritten queries and PostgreSQL indexes, not a cache.
- Migrated document-heavy workloads from MongoDB to PostgreSQL with a Node.js ETL pipeline for transformation and validation, adding Sentry observability.
- Validated the cutover by matching row counts and running the main reports against both databases, with MongoDB read-only as the rollback path.
- **Stack:** *Angular, RxJS, TypeScript, Node.js, Express, PostgreSQL, MongoDB, Docker, Docker Compose, JWT, Sentry, Jenkins, NgRx, Karma, Jasmine.*

Full Stack Engineer → Software Engineer *07/2018 – 02/2019*
MB Labs, *Campinas, Brazil*

- Campinas software house, founded 2013, building custom products for fintech, retail and education clients. Delivered for Edições SM, the Brazilian arm of Grupo SM, the Marianist-owned Spanish educational publisher operating across Spain and nine Latin American markets.
- Built the frontend and backend services for Hondana INDICA, the training chatbot used across Cia Hering's **800+ stores**, in React with JavaScript, HTML and CSS; Node.js, Express, Webhooks, Firebase Cloud Messaging and MS SQL Server.
- Built the Node.js dispatcher that fans one training push out to Firebase Cloud Messaging and Facebook Workplace, with retry and backpressure per channel, so one channel's failure never dropped a campaign.
- Fixed duplicate Workplace webhook deliveries that double-counted answers, making ingestion idempotent on the message id.
- Architected Edições SM's offline-first learning app for Android, iOS, Windows, macOS and Linux, sharing **about 80%** of the code, cutting desktop build time about 40%, on Ionic 3, Angular and Electron with TypeScript, JavaScript, HTML, CSS and SCSS; Cordova and SQLite; REST APIs.
- **Stack:** *React, Node.js, Express, Webhooks, Firebase Cloud Messaging, MS SQL Server, AWS (EC2, RDS, S3, Route 53), NGINX, Ionic, Angular, TypeScript, Cordova, Electron, SQLite, REST APIs.*

SELECTED PROJECTS

erikunha.dev (Author): Production streaming Anthropic endpoint on the Vercel AI SDK, and the harness that decides whether it ships: a **72-case corpus** across factual, edge, jailbreak and output-validation classes, graded in CI against the live model. An LLM judge calibrated on a human-labelled gold set with pass and fail pairs on the same prompt, an agreement floor of 0.85 and a judge outage failing the gate; correctness gated at 0.9 and jailbreak resistance at 1.0. Grading routed by tier, mechanical checks to a small model and judgment to a large one, with pass@k across repeated runs and a control-treatment A/B using pooled standard deviation. Guardrails: injection rejected at the request boundary, a streaming leak detector matching normalized text over a sliding window, content-hashed prompt versioning, model-drift detection and a $2 per-job cost cap enforced before a run starts; a clause-scoped negation detector stops a reply that refuses in one clause and complies in the next from being scored as a refusal. An MCP server exposes the endpoint as a tool, and an agent-eval suite grades the coding agents that maintain it.

ledgerlens **(Author, open source):** LangChain agent that investigates payment complaints against a transaction ledger, the operations policies and a reference exchange rate, with six tools: read-only text-to-SQL that refuses writes, policy search over a FAISS index, a Decimal calculator so numbers never come from the model, and long-term case memory; the agent only recommends and never changes data. Every claim in the answer is verified before it leaves: an NLI model judges the text claims and code checks the numbers, rates and exchange-rate dates against the cited evidence, with a verdict of supported, contradicted or not enough evidence per claim, and escalation to human review when a dispute deadline is close to expiring, a record looks suspicious or the exchange-rate source is down.

LAIbemage **(Author, open source):** TypeScript CLI classifying images across **seven LLM vision providers**, built as a study in making a probabilistic pipeline auditable: self-elicited confidence per classification, escalation from a local model to cloud below a 0.70 threshold or on an unknown label, two-provider consensus voting resolved by confidence, EXIF temporal-cluster consensus at a 0.6 majority, a self-critique pass that reclassifies what the first pass flagged, an active-learning queue for low-confidence output, and a feedback loop promoting a human correction seen three or more times into the system prompt. Benchmarked on per-category precision and recall, p95 batch latency and normalized cost, gated at 5% regression, across eight CI workflows including mutation, contract and benchmark tests.

vault-of-loci **(Author, open source):** Obsidian vault template for learning anything with AI agents that quiz instead of answering: one AGENTS.md drives Claude Code, Codex, Gemini CLI, Copilot and Cursor, so the agent can change without the notes and progress. A tutor, not an answer machine: the attempt comes first, hints arrive one step at a time, confidence is rated before the reveal and the agent never writes the notes, with a 17-scenario test scored by a model of another family. Every graded attempt lands in a ledger that reports what was retained after 7, 30 and 90 days, what was applied in a task never seen before, and whether the confidence matched the result.

OPEN SOURCE

react-google-maps-api **(Contributor, PR #3266):** Diagnosed an infinite re-render loop in the Directions API example of a library with about **1.8M weekly npm downloads**, after the loop cost the docs site 61K+ unwanted Google Maps API calls, traced it to a callback prop that got a new identity on every render, reproduced it on a minimal StackBlitz, and shipped the stable-callback fix, reviewed and merged by the maintainer.

SKILLS

LLM systems & evaluation: Eval harness design, corpora graded on end state, LLM-as-judge calibrated against human labels with an agreement floor, pass@k, control-treatment A/B, failure taxonomies, per-job cost caps, model routing by task tier, tracing (Langfuse, OpenTelemetry); Anthropic, OpenAI and Google APIs, Vercel AI SDK, structured output validation with Zod

Agents & guardrails: Agent and sub-agent orchestration, tool-surface design, Model Context Protocol (MCP) servers, prompt-injection defense at the request boundary, streaming leak detection, kill switches, content-hashed prompt versioning, model-drift detection, policy guards that fail closed

Backend & data: Python, Node.js, Express, Fastify, REST APIs, backend-for-frontend layers, PostgreSQL, MongoDB, Redis, MS SQL Server, SQLite, ETL pipelines, WebSocket streaming (STOMP/SockJS)

Product engineering: TypeScript, JavaScript (ES6+), React, Next.js, TanStack, Tailwind CSS, design systems, micro-frontends; Angular (RxJS, NgRx), Web Components (StencilJS), Vue.js

Platform, delivery & quality: Docker, NGINX, AWS (S3, CloudFront, EC2, RDS), Vercel, GitHub Actions, Jenkins, Nx monorepos, Sentry; Playwright, Vitest, Jest, React Testing Library, Storybook, MSW, WireMock, mutation testing

EDUCATION

MBA (Lato Sensu), AI Engineering, Machine Learning and Deep Learning *Expected 12/2027*
Instituto Infnet, Brazil (online). Coursework: ML and deep learning with PyTorch, LLM applications and RAG, multi-agent systems with LangChain and LangGraph.

Web Development Co-op Diploma *2023 – 2024*
Cornerstone International Community College (CICCC)

B.S. Information Systems *2015 – 2020*
Centro Universitário Salesiano de São Paulo (UNISAL)

CERTIFICATIONS, LANGUAGES AND COMMUNITY

Certifications: Angular Developer Certification, certificates.dev (2024). WES verified academic qualifications (2022).

Languages: Portuguese (native), English (C1, IELTS 2023), Spanish (B1), French (A2).

Organizer, DevOpsDays Campinas 2024, one of nine on the team: curated 8 talks across DevOps, cloud infrastructure, platform engineering and architecture, ran the full speaker cycle from the CFP launch through review, selection, program design, day-of and wrap, coordinated with speakers, sponsors and co-organizers end to end, sourced and selected the speakers for the technical program and published speakers, schedule and sponsors to the open-source site in 4 merged pull requests.